

Periodic Motion:

After a certain time interval, motion is getting repeated.

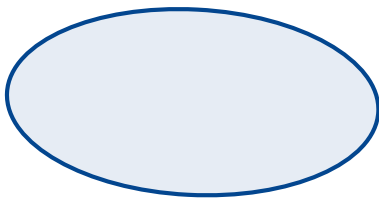
Displacement at time  $t \Rightarrow x(t)$

$$x(t+T) = x(t)$$

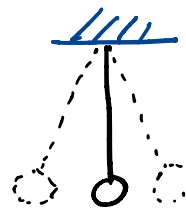
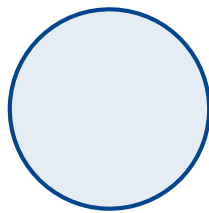
Basic feature

Period =  $T$

Elliptical



Circular



Orbital Periodic Motion

Oscillation / vibration  
Linear Periodic Motion

Oscillation:-

→ Complete oscillation →



→ Period( $T$ ) → time required to make 1 complete oscillation

→ Frequency,  $\nu$  → nr. of complete oscillations in unit time.

$$\nu = \frac{1}{T}$$

→ Amplitude →  
( $A$ )

Mid-point (or) Max displacement at any side



## Mechanics:

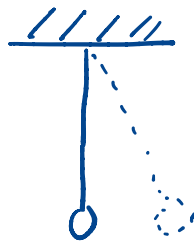
- i) These oscillatory motion has an equilibrium position where no force is applying on them.

When at eq<sup>m</sup> position  $\Rightarrow$  Force = 0  $\Rightarrow$  acceleration = 0

if, at " "  $\Rightarrow$  velocity = 0 No Motion

- ii) Now, if an agent takes them a bit away from their equilibrium position —

- a) Now, a restoring force will start acting on them.



- b) This force will create an acceleration.

- c) due to this, it will reach to the eq<sup>m</sup>. position.

- d) But this time, when it will return to eq<sup>m</sup> position, it will have a non-zero velocity.

- e) This velocity will again take it away from eq<sup>m</sup> position (in the opposite direction).

- f) So, again, the restoring will come into play to take it towards eq<sup>m</sup> position.

$\rightarrow$  This cycle will Repeat.

## Simple Harmonic Motion:

Restoring Force = ??  
(F)

The far away it is from eq<sup>m</sup> position, the stronger its tendency would be to return to eq<sup>m</sup> position.  $\nearrow$

Intuitively,  $F \propto x$

where,  $x \Rightarrow$  displacement from eq<sup>m</sup> position

S, at eq<sup>m</sup> position,  $x=0$

But, force and displacement has to be in opposite direction, physically.

$$F \propto -x$$

Hooke's law.

In words, higher its disp from eq<sup>m</sup>, <sup>proportionally</sup> higher is the force towards eq<sup>m</sup>,

The systems where it's applicable  $\Rightarrow$  Simple Harmonic motion.

$$F \propto -x$$

$$F = -Kx$$

Equation of Motion  
(EOM)

( $K \rightarrow$  Proportionality constant)  
( $K \rightarrow$  Force constant)

## Solving EOM of SHM:

displacement  $\Rightarrow x$

velocity  $\Rightarrow v = \frac{dx}{dt}$

acceleration  $\Rightarrow a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

From Newton's 2nd law,  $F = ma$  ;  $m \rightarrow$  Mass of the particle.

Then,

$$ma = -Kx \quad (\text{EOM also})$$

$$a = \frac{d^2x}{dt^2}$$

$\therefore$

$$m \frac{d^2x}{dt^2} = -kx$$

Differential Eqn of SHM.

Simplify:

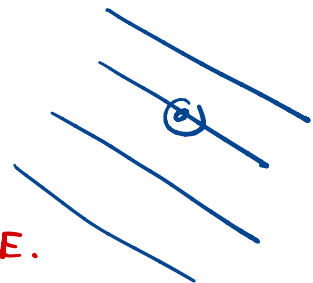
$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\frac{d^2x}{dt^2} = -\omega^2x \quad \left[\omega^2 = \frac{k}{m}\right]$$

Order = 2

Degree = 1 (linear)

Homogeneous (no constant term)



2nd order Linear homogeneous D.E.

↳ 2 Dof (2 integration constants)

↳ It will have 2 Linearly independent solutions.

To curb 2 Dof, we need 2 Independent conditions.  
Let's say 2 initial conditions.

" " , we are told, the displacement at  $t=0$   
and, " velocity at  $t=0$

But, if we have only 1 solution for,  $x = f(t)$

$$\text{then, } v = \frac{dx}{dt} = \frac{df}{dt}$$

So fixing  $x$ , will also fix  $v$ .

But, This should not be the case.

We should be able to start with any arbitrary displacement value and any arbitrary velocity value.



So, 2 linearly independent solutions has to be considered.

Let's say,  $f_1(t)$  and  $f_2(t)$  — are <sup>linearly Independent</sup>  $2_n$  solutions of

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

Then,

$$\frac{d^2f_1}{dt^2} = -\omega^2 f_1 \quad ; \quad \frac{d^2f_2}{dt^2} = -\omega^2 f_2$$

Add them →

$$\frac{d^2f_1}{dt^2} + \frac{d^2f_2}{dt^2} = -\omega^2 f_1 - \omega^2 f_2$$

$$\Rightarrow \frac{d^2}{dt^2}(f_1 + f_2) = -\omega^2 (f_1 + f_2)$$

So  $(f_1 + f_2)$  is also a solution of  $\frac{d^2x}{dt^2} = -\omega^2 x$

Actually,  $f = c_1 f_1 + c_2 f_2$  will also be a solution, if

$f_1$  and  $f_2$  are already solution.  $c_1, c_2$  are arbitrary constant.

$$\frac{d^2f}{dt^2} = \frac{d^2}{dt^2} (c_1 f_1 + c_2 f_2)$$

$$= \frac{d^2}{dt^2} (c_1 f_1) + \frac{d^2}{dt^2} (c_2 f_2)$$

$$= c_1 \frac{d^2f_1}{dt^2} + c_2 \frac{d^2f_2}{dt^2}$$

$$= c_1 (-\omega^2 f_1) + c_2 (-\omega^2 f_2)$$

$$= -\omega^2 c_1 f_1 - \omega^2 c_2 f_2$$

$$= -\omega^2 (c_1 f_1 + c_2 f_2)$$

$$\frac{d^2 f}{dt^2} = -\omega^2 f$$

If,  $f_1$  and  $f_2$  are 2 solutions then,

$$\text{General solution} \Rightarrow f = c_1 f_1 + c_2 f_2$$

These 2 constants would be determined by 2 initial conditions

Let's go back to our case

$$\frac{d^2 x}{dt^2} = -\omega^2 x$$

What can be its solutions?

$$\rightarrow \sin(?t) \quad \leftarrow \quad ? = \omega$$

$$\rightarrow \cos(?t)$$

$$\rightarrow e^{?t} \rightarrow \text{will be in HW.}$$

So, 2 independent solutions  $\Rightarrow \sin(\omega t)$ ,  $\cos(\omega t)$

So, General solution

$$x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t)$$

$\rightarrow$  This is also a feature of SHM.

If 2 conditions are given, using them we can determine  $A_1$  and  $B_1$

$$x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t)$$

What does  $A_1$  and  $B_1$  Physically mean here?

$$x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t)$$

$$v(t) = \frac{dx}{dt}$$

$$= A_1 \omega \cos(\omega t) - B_1 \omega \sin(\omega t)$$

Then, what is  $x$  and  $v$  at  $(t=0) \Rightarrow$

$$x(t=0) = B_1$$

$$v(t=0) = A_1 \omega \Rightarrow A_1 = \frac{v(t=0)}{\omega}$$

$$\text{EOM} \Rightarrow \frac{d^2x}{dt^2} = -\omega^2 x \quad (\omega^2 = \frac{k}{m})$$

$$\text{Solution} \Rightarrow x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t)$$

$B_1 \Rightarrow$  initial position

$A_1 \Rightarrow \frac{\text{initial vel.}}{\omega}$

Another form of solution (more compact)

$$x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t)$$

$$= \sqrt{A_1^2 + B_1^2} \left[ \frac{A_1}{\sqrt{A_1^2 + B_1^2}} \sin(\omega t) + \frac{B_1}{\sqrt{A_1^2 + B_1^2}} \cos(\omega t) \right]$$

$$\text{Now, let's say } \sqrt{A_1^2 + B_1^2} = A$$

$$\text{and, } \frac{A_1}{\sqrt{A_1^2 + B_1^2}} = \cos \alpha$$

$$\begin{aligned}
 \text{Then, } \frac{B_1^2}{A_1^2 + B_1^2} &= 1 - \frac{A_1^2}{A_1^2 + B_1^2} \\
 &= 1 - \cos^2 \alpha \\
 &= \sin^2 \alpha
 \end{aligned}$$

$$\therefore \frac{B_1}{\sqrt{A_1^2 + B_1^2}} = \sin \alpha$$

$$\therefore \frac{\sin \alpha}{\cos \alpha} = \frac{\frac{B_1}{\sqrt{A_1^2 + B_1^2}}}{\frac{A_1}{\sqrt{A_1^2 + B_1^2}}} = \frac{B_1}{A_1}$$

$$\therefore \tan \alpha = \frac{B_1}{A_1}$$

$$\alpha = \tan^{-1}\left(\frac{B_1}{A_1}\right)$$

$$\begin{aligned}
 \therefore x(t) &= \sqrt{A_1^2 + B_1^2} \left[ \frac{A_1}{\sqrt{A_1^2 + B_1^2}} \sin(\omega t) + \frac{B_1}{\sqrt{A_1^2 + B_1^2}} \cos(\omega t) \right] \\
 &= A \left[ \cos \alpha \sin(\omega t) + \sin \alpha \cos(\omega t) \right]
 \end{aligned}$$

$$x(t) = A \sin(\omega t + \alpha)$$

- look we still have 2 independent constants  $\Rightarrow A, \alpha$

What does  $A$  and  $\alpha$  here physically mean?

$$x(t) = A \sin(\omega t + \alpha) = A \sin \theta \quad (\theta = \omega t + \alpha)$$

$$\sin \theta \rightarrow \text{Max val} = +1$$

$$\rightarrow \text{min val} = -1$$

$$\therefore x(t) \rightarrow \text{Max val} = +A$$

$$\rightarrow \text{Min val} = -A$$

$\therefore$  In this solution,  $A$  represents Amplitude of oscillation.

Now, at  $t=0$

$$x(t=0) = A \sin(\omega \cdot 0 + \alpha)$$

$$= A \sin \alpha$$

$$\alpha = \sin^{-1}\left(\frac{x}{A}\right)$$

So  $\alpha$  is basically encodes initial position and Amplitude ratio.

So, among  $\{x(0), v(0), A, \frac{x(0)}{A}\}$  - we can independently fix any 2.

Let's try to find out values & constants

Q1 An agent dragged the particle from the eqn position by 5 units and then left it to move.

Ans.  $x(t=0) = 5$

$$v(t=0) = 0$$

Now,  $x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t)$

$$v(t) = \frac{dx}{dt} = A_1 \omega \cos(\omega t) - B_1 \omega \sin(\omega t)$$

Now,  $x(t=0) = 5$

$$\Rightarrow A_1 \sin(\omega \cdot 0) + B_1 \cos(\omega \cdot 0) = 5$$

$$\Rightarrow B_1 = 5$$

Also,  $v(t=0) = 0$

$$\Rightarrow A_1 \omega \cos(\omega \cdot 0) - B_1 \omega \sin(\omega \cdot 0) = 0$$

$$\Rightarrow A_1 \omega \cdot 1 - B_1 \omega \cdot 0 = 0$$

$$\Rightarrow A_1 \omega = 0$$

$$\Rightarrow A_1 = 0 \quad \left( \because \omega = \sqrt{\frac{k}{m}} \neq 0 \right)$$

Final solution  $\Rightarrow$

$$\therefore x(t) = 5 \cos(\omega t)$$

Q1 An agent dragged the particle from the eq<sup>n</sup> position by 5 units and then left it imparting some velocity so that the amplitude of oscillation becomes 10 units.

Ans.  $x(t) = A \sin(\omega t + \alpha)$

Here, Amplitude =  $A = 10$

Also, at  $t=0 \Rightarrow x(t=0) = 5$

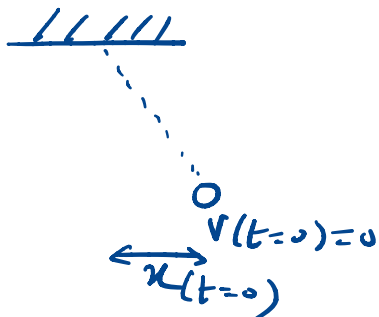
$$\therefore A \sin(\omega \cdot 0 + \alpha) = 5$$

$$\Rightarrow 10 \sin(\alpha) = 5$$

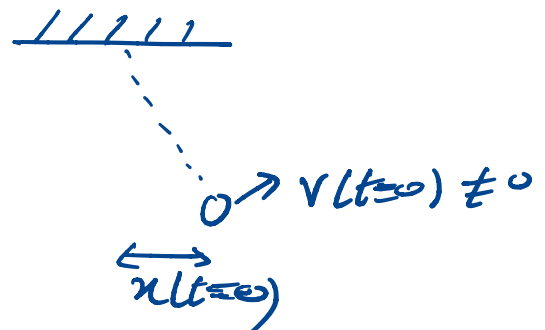
$$\Rightarrow \sin \alpha = \frac{5}{10} = \frac{1}{2}$$

$$\Rightarrow \alpha = \frac{\pi}{6}$$

$$\therefore x(t) = 10 \sin\left(\omega t + \frac{\pi}{6}\right)$$



Then amplitude =  $x(t=0)$



Then, amplitude  $> x(t=0)$

## Relation of velocity and position:

$$x(t) = A \sin(\omega t + \alpha)$$

$$v(t) = \frac{dx}{dt} = A\omega \cos(\omega t + \alpha)$$

$$= \pm A\omega \sqrt{1 - \sin^2(\omega t + \alpha)}$$

$$= \pm A\omega \sqrt{1 - \left(\frac{x(t)}{A}\right)^2}$$

$$v(t) = \pm \omega \sqrt{A^2 - x^2(t)}$$

## Relation of acceleration with position:

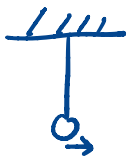
$$a = -\omega^2 x$$

| <u>Quantity</u> | <u>Expression</u>                                                    | abs. value<br><u>minimum</u> | abs. value<br><u>maximum</u> |
|-----------------|----------------------------------------------------------------------|------------------------------|------------------------------|
| $x(t)$          | $A \sin(\omega t + \alpha)$                                          | 0                            | A                            |
| $v(t)$          | $A\omega \cos(\omega t + \alpha)$<br>$= \pm \omega \sqrt{A^2 - x^2}$ | 0                            | $A\omega$                    |
| $a(t)$          | $-\omega^2 x$                                                        | 0                            | $-\omega^2 A$                |

|  | <u><math>x</math></u> | <u><math>v</math></u> | <u><math>a</math></u> |
|--|-----------------------|-----------------------|-----------------------|
|  | +A                    | 0                     | $-\omega^2 A$         |
|  | 0                     | $-\omega A$           | 0                     |
|  | -A                    | 0                     | $\omega^2 A$          |





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